1 January 8, 2026 (Class 0)

SCS CVX Julia DCP - disciplined convex programming SOCP
Norm approximation

2 January 9, 2026 (Class 1)

2.1 Norm Approximation

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$. The objective is

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|$$

Definition 2.1 (ℓ_1 norm): *For $\mathbf{y} \in \mathbb{R}^m$, ℓ_1 norm is defined as*

$$\|\mathbf{y}\|_1 = \sum_{i=1}^{m} |y_i|$$

Definition 2.2 (ℓ_2 norm): *For $\mathbf{y} \in \mathbb{R}^m$, ℓ_2 norm is defined as*

$$\|\mathbf{y}\|_2 = \left(\sum_{i=1}^{m} y_i^2 \right)^{1/2}$$

Definition 2.3 (ℓ_∞ norm): *For $\mathbf{y} \in \mathbb{R}^m$, ℓ_∞ norm is defined as*

$$\|\mathbf{y}\|_\infty = \max_{1 \le i \le m} |y_i|$$

3 January 14, 2026 (Class 2)